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Agent-based modeling to inform flood emergency planning and management

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ABSTRACT

Objective: *Agent-based modeling can provide powerful tools to inform flood emergency management and to provide an assessment of loss of life due to a flood event. The objective of this work is to study the suitability and robustness of this type of models for being applied in practice in managing flood emergencies.*

Design: *This article describes the application of a prototype, agent-based Life Safety Model (LSM) to two populated areas in the Thames Estuary. Parameters sensitivity analyses have also been performed to assess the robustness and the applicability of this model as part of the actual emergency practice.*

Results: *The model of the two areas resulted in the estimation of the number of fatalities for each scenario for different causes such as drowning, exhaustion, building collapse, and vehicles being swept away. The model was also successfully validated against historical data from the 1953 Canvey Island flood.*

Conclusions: *The LSM offers a scientifically robust method of assessing injuries and lives lost, and it allows the comparison of different emergency management strategies that could assist in reducing the loss of life during future flood incidents.*

Key words: flood, disaster management, emergency planning, loss of life, agent based

INTRODUCTION

An accurate understanding of loss of life and injuries as a flood event unfolds is valuable for developing and improving emergency plans for areas at risk. Despite the global impacts of floods, there are a limited number of methods to estimate the loss of life and the evacuation times for flood events. Loss of life modeling can be performed at different levels of detail as follows:

- *Macro or overall event level*, where one mortality rate is applied to the whole of the exposed population.
- *Meso or group/zone level*, where mortality rates are estimated for groups of people or specific zones.
- *Micro or individual level*, where the circumstances and behavior of each individual is modeled to estimate each person's probability of dying.

Until recently, most of the loss of life models for floods were based on a statistical analysis of fatalities and injuries from historical events to date. In the United Kingdom, the work done to assess the loss of life and evacuation times for flood risk areas has been limited to macro- or meso-level estimates. The "Risk to People" model, developed as part of a Department for the Environment Food and Rural Affairs (Defra) research project, is the most commonly used tool in the United Kingdom to assess flood fatalities.[1] However, this meso-level method is based on an empirical, generalized model that does not use detailed information on each individual in its "broad-scale" estimates of loss of life. There are several limitations to empirical approaches to loss of life modeling as follows:

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- Floodwater depths, velocities, and travel times that affect the fate of people and vehicles are based on large-scale average values.

- Factors that change with time are often not represented.

- The population at risk is often considered to be heterogeneous for the entire inundation area or for large subareas of it. Empirical methods do not represent, at a detailed level, the many attributes (eg, age and health) that are important determinants of loss of life.

- Evacuation is not considered as a separate process, and the benefits of those who move to safe havens are often not explicitly included.

- The effectiveness and rates at which the flood warning message is disseminated are not taken into account.

Empirically based loss of life models tend to apply one mortality rate to an area and do not take into account the cause of death. Jonkman analyzed 13 flood events in Europe and in the United States that resulted in 247 reported flood-related fatalities. Each event occurred in the past 20 years and involved relatively few deaths. Jonkman considered events to be representative for extreme events in most Western European countries.[2] Drowning accounted for some 67 percent fatalities. Vehicle-related casualties for drowning were found to occur most frequently as a result of people attempting to drive across flooded bridges, roads, or rivers. Physical traumas account for 12 percent of the fatalities, most of which occurred while people were in their vehicles. Other causes of death included heart attacks during evacuation and return, electrocution during clean up, and deaths from fires following a floods.

To provide a more accurate assessment of loss of life and evacuation times, an agent-based model is required. An agent-based model is a computational model that simulates the interactions of autonomous receptors with a view to assess their effects on the system as a whole. It can model the simultaneous operations of multiple "agents" or receptors (in this case, people and vehicles) with the floodwater, in an attempt to re-create and predict the actions of complex phenomena such as those that occur in flood emergency. The modeling of the evacuation process generated by an approaching flood is included within such a model. This is important for those responsible for flood event management planning. It can identify "bottlenecks" in the escape network before they are experienced in an evacuation, it can also be used to determine the impact of road closures due to flooding, the impact of phased evacuation on traffic loading, and many other possible consequences of an evacuation event. In the United Kingdom, there has been little work undertaken for evacuation modeling specifically for flood event management.

The objective of this work is to study the suitability and robustness of agent-based models for being applied in practice in managing flood emergencies.

In particular, this article describes the application of a recently developed, agent-based, microlevel Life Safety Model (LSM). This work was undertaken as part of Task 17 of the European Community funded research project FLOODsite.[3] Rather than relying on very scarce and possibly unrepresentative observations on life loss caused by large floods, the LSM is designed to generate insightful information about this complex phenomenon by observing the simulated behavior of a physically based virtual representation of the inundation area and its inhabitants as they mobilize to escape flooding. The LSM allows the comparison of different emergency management strategies (eg, the use of safe havens) that can assist in reducing the loss of life during future floods and dam breaks.

THE BC HYDRO LIFE SAFETY MODEL

Background

The LSM is a piece of prototype software developed by BC Hydro in Canada that previously had only been used to carry out dam break risk assessments for small communities (eg, less than 3,000 people) in Canada. The LSM allows dynamic interaction

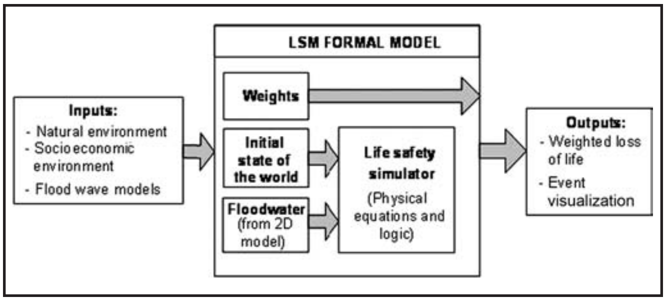

Figure 1. High-level architecture of the Life Safety Model (Source: Refs. 4 and 5).

between the receptors (eg, people, vehicles, and buildings) and the flood hazard. The LSM requires a significant amount of data including:

- the location of individual properties, vehicles, and people;
- flood depths and velocities from a two-dimensional hydrodynamic model of the flood event; and
- details of the road network and other pathways.

Figure 1 provides a conceptual view of the architecture of the LSM. The core of the system is the LSM Simulator that requires two inputs: an initial state of the world (which describes modeling receptors such as people, buildings, cars, and roads) and the flood wave. The simulator output includes an estimated loss of life and dynamic computer graphics visualizations.

The system models the "fate" of a set of receptors, which are described by their position at each time step through the simulation. Each receptor can have a set of properties that describes its normal location/condition during a week, such as travel times, school/work hours, and weekend activities. Other time-varying properties include the ability of the people, vehicles, and buildings (ie, the receptors) to withstand the effect of the floodwater. The LSM also models how people would react to the approaching wave, with and without a formal evacuation warning. Before any loss-life-modeling can be carried out, the following has to be undertaken:

- Two-dimensional hydrodynamic modeling of the flood area to produce a temporal varying of grid of depth and velocity.
- A "virtual world" needs to be set up representing the initial state of the world before the flood occurs.

The representation of the population at risk and their associated vehicles, buildings, and other infrastructures are encompassed in what is termed a "People's World." The various features that input to and affect on this world, such as the flood wave and the flood warning systems, are realized via a set of submodels that can collectively operate on the People's World. Figure 2 shows representation of the People's World in the LSM. At the beginning of a simulation, all the people at risk are either located in a building or in a vehicle on the road network. A number of different People's Worlds can be set up depending on the time of day, day of the week, and the time of the year.

The LSM uses a generalized event logic to determine the location of each person, whether they are aware of the flood wave, whether they are trying to find a safe haven, what happens if they encounter the flood, and whether they survive or not. A loss function related to each receptor (eg people, buildings, and vehicles) specifies the ability of a receptor to resist the impact from the flood wave, in terms of depth and velocity, and how these can change during an event. There can be instantaneous loss when an individual encounters with fast-flowing water, a group who has sought safety in a building can suffer cumulative loss if the building collapses, or a slow deterioration in health if they are exposed to the flood water for a significant length of time, as a result of hunger or cold.

As a flood event evolves, the interaction of receptors with the flood wave will impact the ultimate loss of life. The timing of the event and the decisions made by individuals can determine whether or not they can escape the flood wave. As the flood progresses, escape routes can be eliminated by rising water, and with advancing time roads can become congested with evacuees.

The internal logic of the LSM can be explained by considering how an individual might experience a flood

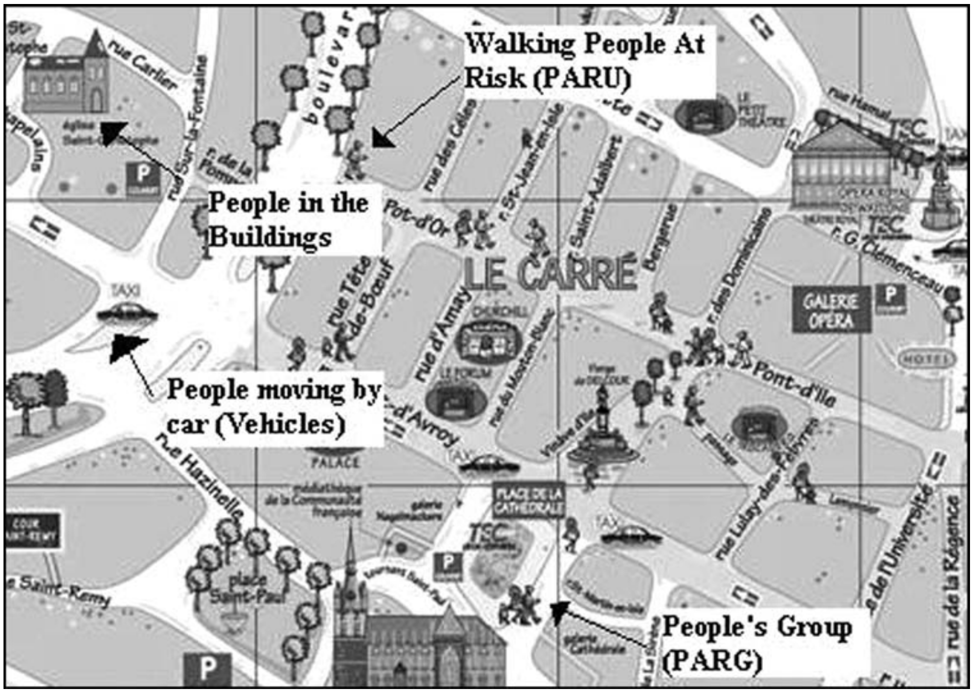

Figure 2. Representation of the "People's World" in the LSM. PARU is a "People at Risk Unit," which is an individual person as modeled by the LSM. PARG is a "People at Risk Group," which represents a group of people constituted by two or more PARUs.

event. Figure 3 shows a person located in a building at the start of a flood event. Assuming that the area will be heavily inundated by floodwater, the person would be killed if caught in the building without warning in the location denoted by "A" in Figure 3. Three possible safe havens are shown to which the person can evacuate on foot or in a vehicle. Taking into account the

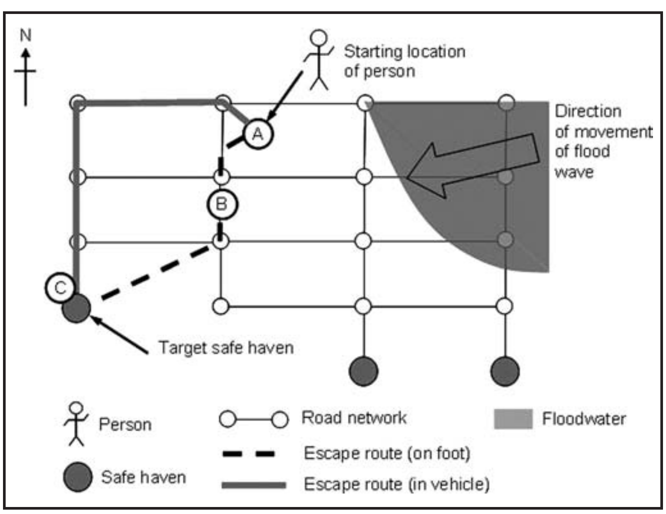

Figure 3. Fate diagram for a person in the LSM (Source: Ref. 6).

"costs" to reach each haven, the south-west alternative is optimal for both foot and vehicle escapes. However, if the person attempts to escape on foot, they will be overwhelmed at point B. Under the third scenario, the person survives due to a combination of sufficient warning and the use of a vehicle to reach point C. The ability to generate and assess the outcomes of multiple scenarios is a key capability of the LSM.[6]

Application of the LSM in a UK environment

As part of the research, work was carried out to test the LSM in a UK environment. The aims of testing the LSM in the United Kingdom were as follows:

- To assess the possibility of employing the LSM for dam risk assessments and flood event management planning, rather than its original purpose of emergency planning for dam breaks.

- To assess whether it was possible to use the prototype version of the LSM in the United Kingdom to estimate evacuation times for people using data readily available in the United Kingdom.

- To test if the LSM could be applied to 40,000 individual receptors in the United Kingdom. This number of receptors is an order of magnitude more than it had been applied in the past.

- To see whether it was possible to compare the results of the LSM in terms of evacuation times with other evacuation models developed for the project.

- To assess the accuracy of the estimates of loss of life and building collapse provided by the LSM.

APPLICATION OF THE LSM TO CANVEY ISLAND

Background

Canvey Island, shown in Figure 4, is an island in the Thames Estuary, covering an area of 18.5 km^2. The mean high water mark of the Thames Estuary at Canvey Island is higher than most of Canvey Island's land. The first sea defenses were constructed in 1623, and the Dutch settlers formed the first Canvey Island communities of the modern era. The population did not expand rapidly until the 1920s, with 1,795 inhabitants in 1921, but the population expanded more than 6,000 in 1927. In the latter period, the number of buildings rose from 300 to about 1,950.[7] In 2001, its population was estimated to be approximately 37,000.[8]

In 1953, the island was inundated by the "Great North Sea Flood" that breached flood defenses and resulted in the deaths of 58 people and the destruction of several thousand houses.[9] The likelihood of flooding of the access routes to and from Canvey Island will increase following the rise in sea level. Access to Canvey Island is currently possible only by two roads, both of which are connected to the same roundabout. Any disruption to these routes would hamper evacuation and severely limit access.

At present, Canvey Island is protected by a concrete sea wall that rises approximately 3 to 4 m above the high tide level. However, it has been found that while substantial, these defenses show signs of deterioration such as cracks in the concrete and the degradation of seals between slabs.[9] Although the current standard of protection at Canvey Island is of 0.1 percent (1 in 1,000 years), this will be reduced to 0.5 percent (1 in 200 years) by 2030 owing to the rise in sea level and the subsidence affecting the south of the United Kingdom.

In Canvey Island, it has been estimated that 30 percent of properties are bungalows and 45 percent of

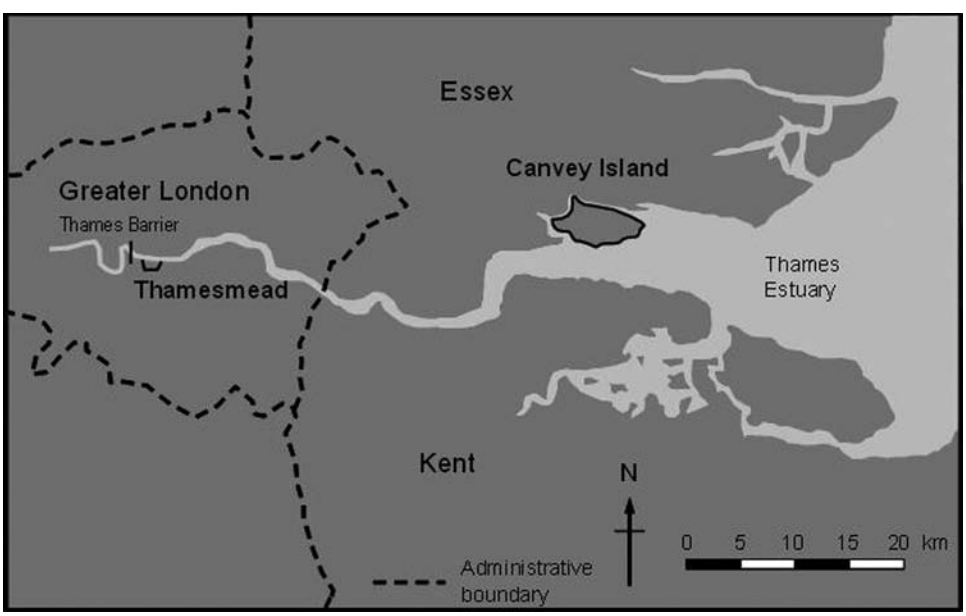

Figure 4. Locations of Canvey Island and Thamesmead.

flats are situated at ground floor level, and thus, there is a large risk to life and property with limited opportunities to temporarily move to a higher level.[9] It is possible that a majority of the island would be inundated if a major storm surge occurred and led to major overtopping or breaching of defenses.

Available data

One of the key tests of using the LSM in the United Kingdom was to assess whether there was sufficiently readily available data to use the model. The readily available data for Canvey Island comprised the following:

- Population data were available from the Office for National Statistics at an output area level. Output areas contain an average of around 125 houses.
- Number of vehicles at an output area level is available from census data.
- Topographic data in the Thames Estuary LIDAR survey data were available with a vertical accuracy of approximately ±25 mm.
- The locations of properties were available in the form of a national property dataset that provides geo-referenced details of each of the properties in England and Wales.
- The road network was digitized from street and Ordnance Survey maps.

These data were used to construct a "virtual" representation of the modeled areas that were used by the LSM. The majority of the effort expended in the work was related to setting up the virtual representation of the areas of interest.

Hydrodynamic modeling of the 1953 flood

To model the 1953 Canvey Island flood, a historical analysis was undertaken to reconstruct the situation that existed at that time. Important sources of information included historical maps of the island, articles from 1953 newspapers, books,[7] police reports, and the results of physical modeling carried out in 1954. This information was used to assist in assessing the height and location of the 1953 flood defenses, to update the digital terrain model, to reconstruct the tidal shape, and to assess the incoming flood volume associated with the breaches that occurred.

The modeling technique used here is finite difference model provided by the software TUFLOW. Being a finite difference modeling, the domain has been discretized in a regular 20-m grid, covering the total domain with an area of 18 km^2. The time span of the model is 8 hours.

The results of the hydrodynamic model indicated that the 1953 flood covered most of the Canvey Island. The model showed that the water depth was 3 to 4 m at the point closest to the breach with a mean depth of between 0.8 and 1.0 m. The modeled volume of the 1953 flood was estimated to be 12.6 million m^3. This compares well with a 1953 flood volume for Canvey Island of 11.7 million m^3 that was estimated by the Kent and Essex River Board shortly after the event.[10]

Results for the 1953 flood

The results of the reconstruction of the 1953 flood event agreed well with the available historical data. The BC Hydro LSM model indicated that approximately 100 to 120 fatalities had occurred during the 1953 event. This number is dependent on the "resilience factors" applied to both people and buildings. The actual number of people who died in 1953 was 58. The number of buildings destroyed during the event is unclear. However, the available anecdotal evidence seems to be similar to the LSM model results in the order of magnitude, even if this number of destroyed building in the LSM seems to be overestimated. Figure 5 shows the results of the loss of life modeling using the LSM for the 1953 flood. Of the 13,000 people living in the island in 1953, the model indicated that there would be 119 fatalities, 48 as a result of drowning and 71 as a result of exhaustion. The LSM also indicated that some 2,300 would be "toppled" (ie, knocked over) by the floodwater. This

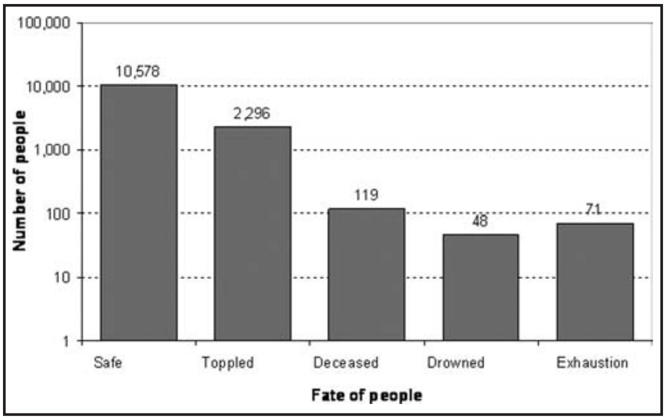

Figure 5. Typical results of modeling the fate of people for the 1953 Canvey Island flood.

figure can be used as a proxy for the number of injuries that are likely to occur.

Parameter sensitivity analysis

The 1953 Canvey Island modeling has been a good workbench to test the sensitivity of the model to the variation of the model's parameters.

Rate of warning. In the model, the velocity of the alarm dissemination, starting from the warning center, can be set up. This parameter states the rate of the warning spreading out from the warning center in the radial way. The model is sensitive to the variation of this parameter. It has been noted that the number of deceased people is 10 times bigger if the parameter is divided by three, but it also obviously depend on other parameters such as the position of the warning center. Therefore, the sensitivity in this parameter is fully justified with the analysis of the real world. In fact, in different case studies, it has been proved that the number of fatalities varied in a significant way depending on the velocity in the dissemination of the alarm.[11]

Position of the warning center. The physical place in which to put the warning center is really important and can strongly affect the simulation. Usually, it is clear where the warning center can be situated (as the police or fireman station), but the way of spreading the warning is unclear. Therefore in the LSM, the warning center can be modeled only as a point from which the warning is spread radially. This means that it can be assumed

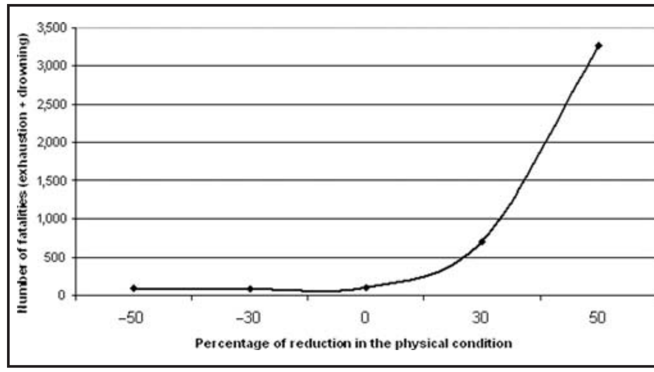

Figure 6. Variation in the number of fatalities with the decrease in the people's physical condition.

that the warning is disseminated by means of door-by-door alert by officers initially located in the warning center. For that, in the 1953 model, the warning center has been located in proximity of the police station.

Proximity warning distance. This parameter represents the minimum distance between two people that allows them to warn each others. The definition of this distance is a bit tricky because it might depend on different factors that are mainly of social aspects. Furthermore, the model has revealed to be very sensitive to this factor. For example, the fatalities increase from 82 to 201 while decreasing the proximity warning distance from 150 to 40 m.

Initial physical condition. This value states the initial condition of the people. This parameter describes the physical strength of the modeled person in relation to his\her capacity to survive if exposed to the flood wave. This is one of the most sensitive parameters of the model. It has been noticed that the death for exhaustion increases from 50 to 674 people just decreasing that parameter of 30 percent. A graph of the variation in the number of fatalities with the decrease in the people's physical condition is shown in Figure 6. The detailed results are shown in Table 1. It can be noted from the table that the variation of the physical condition mainly affects the number of fatalities due to exhaustion. In fact, even if the initial physical condition partly affects the drowning threshold, the physical condition parameter decreases with the time, and thus, the lower is the initial value, the shorter is the time in which someone can be exposed to the flood.

Table 1. Number of fatalities calculated against the reduction of the people's physical condition parameter (PPC)

Percentage reduction in PPC	No. of people "toppled"	No. of fatalities	
		Fatalities due to drowning	Fatalities due to exhaustion
−50	102	55	36
−30	107	55	39
0	130	54	50
30	177	37	674
50	158	55	3,218

Buildings structural condition. The model has not been shown to be particularly sensitive to the variation of the building structural parameter. Therefore, this parameter is very important because in all of the modeling that has been performed, it seems that the number of the destroyed building is overestimated, even though no evidence of the actual number of destroyed buildings in 1953 has been found.

Evacuation time. The time in which the population reaches the exit of the island is obviously dependent on different factors. One of the factors is the escaping speed of foot. This is dependent on the physical condition of the people and decreases with that, ie, the lower is the people's physical condition, the slower they escape. This affects not only the number of fatalities but also the percentage of people that reaches the exit after 8 hours, as is shown in Figure 7.

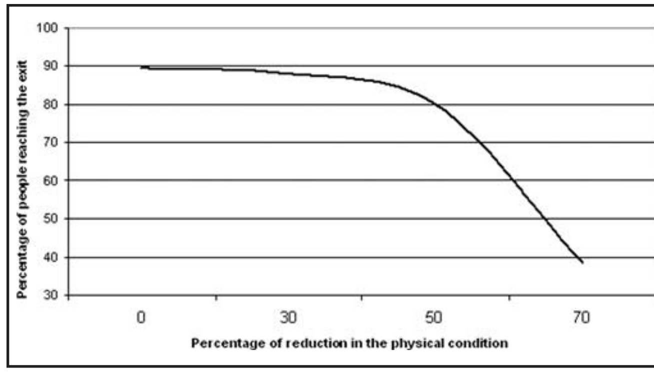

Figure 7. Variation in the percentage of people reaching the exit after 8 hours (net of the number of fatalities) with the decrease in the people's physical conditions.

APPLICATION OF THE LSM TO THAMESMEAD

Having validated the LSM on Canvey Island, it was also applied to the Thamesmead embayment located downstream of the Thames Barrier (Figure 4), which has a population of 43,000. Sixty different scenarios were modeled for the Thamesmead embayment. These included different rates of warning; numbers of road closures and safe havens. For the 60 different scenarios modeled, the number of fatalities varied from a minimum of 406 to a maximum of 2,378 people. The average number of fatalities was found to be 1,296. There are approximately 43,000 people who are exposed to the flooding in the Thamesmead embayment, and thus, the LSM model indicates that on average about 3 percent of the exposed population will suffer fatalities.

Research by Jonkman indicated that the expected number of fatalities is usually between 0.7 and 1.3 percent of the exposed population. However, in these cases, many of the population have evacuated before the hazard occurred. For Thamesmead, the "worst case" of everybody being at home was assumed. In the historical data collected by Jonkman, many of the people had already been evacuated from the exposed area, and therefore, it is expected that in the case of Thamesmead where it was assumed that no evacuation would occur prior to the flood event, the percentage of fatalities would be much higher.

The LSM indicated that the time required for 43,000 people to evacuate the Thamesmead embayment varied between approximately 5 and 8 hours, depending on the number of safe havens assumed and the capacity of the road network. These evacuation times were compared against evacuation times estimated using other methods and were found to be realistic.

CONCLUSIONS

The LSM is a powerful tool that combines the results of the hydrodynamic modeling with the "element at risk" in the domain. This is very useful to give back the first clear picture about the development of a flood event in a populated environment.

The LSM model is the only model that has a dynamic interaction between the receptors (eg, people and vehicles) at risk and the flood hazard. Generally, other loss of life and evacuation models provide only first order of magnitude in terms of the evacuation times and fatalities. These could be useful at high-level planning stage but are unlikely to be useful for detailed emergency planning

The LSM offers a scientifically robust method of assessing residual risk behind flood defenses and downstream of dams in terms of fatalities. The results of the LSM represent the full range of consequences of a flood. In fact, this tool not only computes loss of life but also gives an estimation of the injuries and the damages to the objects such as buildings and vehicles losses. Therefore, the model requires a large amount of data for the simulation setup, which can be consequently very time consuming. Furthermore, the analysis reveals that the model is clearly sensitive in the choice of parameters, as to the rate of alarm disseminating, and to the distance in which the people can communicate the warning between each others. On the other hand, the model sensitivity to each variable is coherent with the sensitivity of the real world on the same variables. For instance, the number of injuries is related on the velocity of warning dissemination that is also strictly related on the methodology with which the population is warned.[11] However, for example, because of the high sensitivity in the physical condition of the people, the choice of this parameter is particularly critical. For this, it would be ideal to use the model not as an absolute estimate of the life losses but as comparison between different strategies and emergency management techniques.

In fact, the LSM allows the comparison of different emergency management strategies (eg, the use of safe havens) that can assist in reducing the loss of life during future floods and dam breaks. The model was validated against historical data from the Canvey Island flood in 1953, during which 58 people lost their lives. The LSM was then applied to Thamesmead to estimate loss of life and evacuation times for a range of scenarios. It is crucial that emergency planners, managers, and responders, as well as flood risk managers, use the best reasonably obtainable scientific information to assess risks to health, safety, and the environment to improve emergency planning and minimize loss of life.

Agent-based models are rarely applied to improve the emergency planning for flood events and to inform flood emergency plans. This article demonstrates that there are potential benefits for flood risk managers to incorporate this type of modeling into their flood event management work and demonstrates the feasibility of the use of scientifically robust agent-based models to improve emergency planning for floods.

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